Bearings

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Bearings Practice: The bearing of a buoy from a harbour is 118°. Find the back bearing of the harbour from the buoy.

Answer: Since 118° is less than 180°, add 180°: 118° + 180° = 298°. The back bearing is 298°.

Bearings Basics (15 questions)

1. What is the bearing of South, as a three-figure bearing?
2. What is the bearing of West, as a three-figure bearing?
3. Write 5° as a three-figure bearing.
4. Write 75° as a three-figure bearing. A) 7.5 B) 75 C) 075 D) 750 E) 0075
5. A boat sails on a bearing of 000°. In which compass direction is it travelling?
6. What is the bearing of North, as a three-figure bearing?
7. A car travels on a bearing of 270°. In which compass direction is it travelling? A) North B) East C) South D) West E) South-West
8. List the 8 compass directions in anticlockwise order starting from North.
9. What is the approximate bearing of South-West (SW)?
10. A cyclist is told to head on a bearing of 045°. Which compass direction does this match? A) North-East B) South-East C) South-West D) North-West E) East
11. Select ALL that apply: which of the following are valid three-figure bearings? A) 090° B) 12° C) 005° D) 400° E) 299°
12. What is the bearing of East, as a three-figure bearing?
13. A plane flies on a bearing of 000° and then turns to fly on a bearing of 180°. What is the relationship between these two directions?
14. Select ALL that apply: which of these bearings point in a generally northerly direction (less than 090° or greater than 270°)? A) 030° B) 100° C) 200° D) 320° E) 355°
15. A compass needle points on a bearing of 180°. Which compass direction is this? A) North B) East C) South D) West E) South-East

Reading & Finding Bearings (15 questions)

1. A ship's radar shows an island at a bearing of 220° from the ship. Describe, using a compass direction, roughly where the island lies relative to the ship.
2. The bearing of village B from village A is 035°. What is the back bearing (the bearing of A from B)?
3. The bearing of a tower from a car park is 198°. What is the back bearing of the car park from the tower?
4. A bearing of 080° is given. What is the back bearing? A) 240° B) 250° C) 260° D) 270° E) 280°
5. A boat sails from harbour P to harbour Q on a bearing of 155°. What is the bearing from Q back to P?

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6. Select ALL that apply: which of these pairs correctly show a bearing and its exact back bearing? A) 010° and 190° B) 100° and 190° C) 215° and 035° D) 330° and 150° E) 060° and 260°

7. A bearing of 112° is measured from North, turning clockwise. Through how many degrees would you turn, starting by facing North, to face along this bearing?

8. Two signposts show that Oakfield is on a bearing of 302° from Mill Lane. What direction, roughly (to the nearest compass point of N, NE, E, SE, S, SW, W, NW), is Oakfield from Mill Lane? A) North B) North-East C) North-West D) West E) South-West

9. A bearing of 340° is given. What is the back bearing?

10. A plane flies on a bearing of 162° from airport X to airport Y. Describe the back bearing in words (using a compass direction) and give its value as a three-figure bearing.

11. A bearing of 005° is given. What is the back bearing? A) 175° B) 180° C) 185° D) 190° E) 195°

12. Select ALL that apply: a ship travels from a buoy to a jetty on a bearing of 205°. Which statements about the return journey (jetty to buoy) are true? A) The back bearing is 025°. B) The back bearing is found by adding 180° to 205°. C) The back bearing is found by subtracting 180° from 205°. D) The return journey is on a bearing of less than 090°. E) 205° and 025° differ by exactly 180°.

13. A bearing from a ship to a buoy is measured as 325°. What is the bearing from the buoy back to the ship?

14. On a map, a farmhouse is due South-East of a barn. Give the bearing of the farmhouse from the barn.

15. A diagram shows two points, E and F. The bearing of F from E is 088°. State the bearing of E from F.

Standard / Advanced (15 questions)

1. A plane flies from city P to city Q on a bearing of 096°. What bearing should it fly to return directly from Q to P?

2. A hiker walks from a car park to a viewpoint on a bearing of 205°, then walks straight back to the car park. What bearing does the hiker travel on for the return journey?

3. Two walkers leave the same campsite at the same time: Walker 1 on a bearing of 050°, Walker 2 on a bearing of 170°. What is the angle between their two paths?

4. A ship sails on a bearing of 110° for 5 km, then turns through 90° clockwise and sails a further 4 km. What bearing is the ship now sailing on? A) 180° B) 190° C) 200° D) 210° E) 220°

5. A surveyor measures the bearing from point X to point Y as 162°, but the reading was actually 20° too small. What is the corrected bearing from X to Y, and what is its back bearing?

6. Select ALL that apply: the bearing from a house to a shop is 240°. Which of these statements are true? A) The back bearing (shop to house) is 060°. B) The shop lies roughly to the south-west of the house. C) The back bearing is found by subtracting 180° from 240°. D) The house lies roughly to the south-east of the shop. E) The angle between the bearing and its back bearing is 180°.

7. A ship sails 9 km on a bearing of 040° from point A to point B. A second ship sails from point A on a bearing of 310° for 6 km to point C. What is the angle ∠BAC (the angle at A between the two ships' courses)?

8. A plane flies from airport A on a bearing of 115° to airport B, then from airport B on a bearing of 295° to airport C. What can you conclude about the positions of A and C? A) They are the same point. B) C is north of A. C) C is south of A. D) There is not enough information to say. E) A and C cannot both exist.

9. A lighthouse keeper notes that a ship is on a bearing of 068° from the lighthouse. What bearing would the ship's captain record for the lighthouse, as seen from the ship?

10. Select ALL that apply: a plane's bearing from Town M to Town N is 134°. Which of the following are correct ways to describe the return bearing from N to M? A) 134° + 180° = 314° B) 314° C) 180° − 134° = 46° D) The return bearing and the outward bearing differ by 180°. E) 134° − 180° = −46°, so add 360° to get 314°.

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11. A ship on a bearing of 316° from a harbour turns and heads directly back towards the harbour. What bearing is the ship now on, and which compass direction is this closest to? A) North B) North-East C) South-East D) South E) South-West

12. A bearing of 198° is given. Find the back bearing, and state which of the 8 compass points the back bearing is closest to.

13. A ship travels on a bearing of 112° for 9 km to point B, then on a bearing of 292° for 4 km to point C. Explain what this tells you about the position of C relative to the ship's original starting point, and find how much distance the ship could have saved by sailing directly from the start to C instead of travelling via B.

14. The bearing from a cabin to a jetty is 4x°, and the back bearing from the jetty to the cabin is (x + 210)°. Form an equation using the fact that the back bearing is 180° more than the outward bearing, and solve for x.

15. A plane's outward bearing from airport R to airport S is (y + 20)°, and the back bearing from S to R is (3y + 10)°. Using the fact that the back bearing is 180° more than the outward bearing, form an equation and solve for y.

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